

Estimating the Jobs Impacts of Different Carbon Tax Reforms

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Introduction

There is broad consensus that sufficiently high carbon prices are important in moving countries to low-carbon pathways. Carbon pricing disincentivizes carbon-intensive activities, and the revenues generated could finance climate actions as well as support households and firms. When climate policies – such as renewable energy investments and energy efficiency programs – are designed in a way that stimulates job creation, a double dividend can be achieved: lower greenhouse gas (GHG) emissions and more jobs. The analysis reported here assesses different policy designs around a carbon tax reform, focusing on the direct and indirect implications for jobs. By filling this important knowledge gap, governments will be empowered to take better-informed and possibly more ambitious action.

It is important for many governments to compare expected economic outcomes with outcomes in other countries, which they consider their peer, and to understand better why results might differ. Economic outcomes can be compared using several indicators. Often, governments want to compare with their regional neighbors, for instance, to find out if investment in clean energy and a carbon tax have the same effects in Egypt and Morocco. As countries in a region often differ in economic progress and income level, another interesting comparison might be within the same income group. More aspirational would be a comparison with a member of the next higher income group. Physical endowment, especially with fossil fuels, can influence the resulting employment effects under different scenarios. Hence, a comparison across producers of crude and processed fuels is suggested in this report.

The possibility space is mapped with different scenarios, applying identical carbon taxes and revenue recycling schemes to all countries to allow for cross-country comparisons. After this introduction, a brief literature review provides the background for the policy scenario designs. It explains the benefits of carbon taxes, as well as the expected impacts of different revenue recycling schemes. The scenarios are described in more detail in section 4, preceded by a brief description of the modeling approach and a short section on caveats regarding the approach in sections 4 and 5. Section 6 explores the results along several dimensions, such as geographic regions, country income groups, and resource endowments, and shows the results disaggregated into 10 economic sectors. Section 6 concludes with an outlook on future research.

Carbon taxes, revenue recycling, and clean energy investment

Price-based policy measures aimed at curbing greenhouse gas emissions encompass various strategies such as carbon taxes and cap-and-trade systems. Carbon pricing assigns a monetary cost to greenhouse gas emissions, incentivizing individuals and businesses to reduce fossil energy consumption. With carbon taxes, governments set explicit prices for carbon emissions, resulting in increased costs of fossil fuels. Revenue generated from carbon taxes can be used to fund various initiatives aimed at mitigating the impact of climate change. Under Emissions Trading Schemes (ETS), governments establish a maximum quantity of emissions and issue allowances accordingly. The companies covered by ETS trade these allowances, determining the price of carbon emissions through market mechanisms. However, ETS is highly administratively challenging and difficult to implement unilaterally due to carbon market liquidity problems, and their fluctuating prices hamper economic planning. Hence, this paper analyzes carbon pricing through taxation.

Notably, carbon pricing may need to be part of a broader climate policy package to reduce GHG emissions effectively. Evidence, mostly from high-income countries, suggests that existing carbon pricing schemes have led to minimal GHG emission reductions: A meta-analysis of empirical studies shows that the carbon pricing schemes analyzed, on average, induced 0 percent to 2 percent reductions in the jurisdictions' annual emissions (Green 2021). This remains below IPCC estimates of GHG emissions needed to fall by 45 percent below 2010 levels to limit warming to 1.5°C (IPCC 2018). Carbon pricing schemes seem not to have induced low-carbon investment or innovation (Lilliestam, Patt, and Bersalli 2021). Moreover, existing carbon prices are well below the price levels needed to achieve the Paris Agreement goals (cf. Stiglitz and Stern 2017). Hence, carbon pricing needs the support of other policy instruments to increase investment in renewable energy and efficiency.

Carbon pricing has been shown to offer a range of economic benefits beyond reducing GHG emissions. By encouraging behaviors that reduce energy consumption and promote the adoption of clean energy, carbon pricing supports global efforts to combat climate change and efforts to improve local air quality (Sampedro et al. 2023). Furthermore, carbon pricing mechanisms provide political certainty about future increases in fossil fuel prices for investors. Carbon pricing can thus play a role in the innovation and adoption of low-carbon technologies, fostering investment in sustainable energy solutions (Calì et al. 2022). Finally, carbon taxes can generate significant revenue streams for governments. The receipts can be directed towards supporting further climate mitigation or development priorities (e.g., funding public expenditure initiatives on education or health) or supporting jobs-rich growth. Carbon taxes are hard to evade, so they increase fiscal space for overall budgetary uses or targeted measures.

However, fiscal reforms entailing carbon taxes also come with certain challenges, including public acceptance. One of the primary political concerns surrounding policies for low-carbon transitions is the level of public acceptance and perceived equity. Transitioning towards resilient and low-carbon structures has the potential to yield favorable impacts on overall employment, extending beyond just the energy sector, as highlighted by recent research (Marin and Vona 2019; Metcalf and Stock 2020; Schoder 2022). However, of particular concern is the heterogeneity of labor market responses and employment outcomes across different occupations, regions, income levels, and genders, which are still not fully comprehended. For consumers, higher fossil fuel prices resulting from carbon pricing may reduce household purchasing power and impact living standards. Similarly, energy-intensive industries may face

challenges in maintaining their competitiveness due to increased production costs, potentially leading to job losses and reduced industrial output.

The revenue generated from carbon taxes can be used strategically to address existing economic challenges and support sustainable development goals. This includes direct income support to low-income households, funding job retraining programs for affected workers, and investing in low-carbon technologies. Revenues can also be spent on public infrastructure and welfare programs (Timilsina 2022). Furthermore, revenues can substitute existing distortionary taxes, mainly, income tax and indirect taxes. Reducing such distortionary taxes can lead to better economic outcomes in terms of GDP and welfare, compared to absorbing carbon revenues in government budgets or providing direct income support to households (Goulder 2013).

Shifting energy investment towards renewable options leads to more jobs and growth at the country level. Evidence on such renewable energy benefits is broad, based on different modeling and empirical approaches, mostly for developed countries and under various scenarios. The aggregate results often show either close to no effects on jobs, meaning job losses net out job gains, slightly positive effects (for instance, for Europe, see (Barker et al. 2016)), or significantly positive effects. The literature agrees that net effects need to be studied, and job losses in high-emitting sectors are counterbalancing the positive effects of investing in clean energy. Such effects are even larger at subnational levels, but we did not go below the national level in the current analysis. Countries specializing in manufacturing renewable energy technologies gain from trade and increased export opportunities, while countries depending on fossil fuel imports gain from trade effects by becoming less dependent when moving to alternative energy sources. At the EU level, transformation scenarios are also assessed in terms of their socioeconomic consequences by soft-linking energy system models with macroeconomic models (Fragkos, Fragkiadakis, and Paroussos 2021, Vrontisi et al. 2020). Nieto et al. (2020) analyze different pathways and find positive effects on employment. Other modeling results include Pai et al. (2021), who find an overall increase of energy jobs globally of 7.7 million by 2050, using an integrated assessment model augmented with country-specific labor productivity data.

The above-listed revenue effects can enhance or alleviate the effects of the clean energy transition in general. More general effects include budget effects on fossil fuel importers and exporters. For the importing countries, GDP will be positively affected by a better trade balance, with renewable energy and energy efficiency options fueled by domestic resources and lowering the need for imported fuels. Oil exporters will see less export demand, reducing returns from royalties. Emission reduction reduces health risks, contributing to the overall target of mitigating climate change. These effects are independent of the choice of mitigation policy instruments and tax design.

The evidence for developing countries is still nascent but growing. Both the impacts of carbon taxes and additional investment in renewable energy have important implications for developing countries. With low tax incidence and tight public budgets, carbon taxes can increase the fiscal space and channel revenues toward achieving sustainable development goals. With roughly 675 million people globally without electricity access,¹ 2.3 billion people without access to clean cooking, and up to 3.5 billion households without access to reliable and affordable energy solutions (Ayaburi et al. 2020), investing in

¹ <https://www.irena.org/Publications/2023/Jun/Tracking-SDG7-2023>

renewable energy provides a solution not only for the climate challenge, but also towards better economic opportunities, better jobs, and growth. Relatively few studies have considered the specificities of job implications in lower-income countries when analyzing the impacts of carbon prices, for example, accounting for informal labor or remittances. The World Bank's Country Climate and Development Reports (World Bank Group 2022) give an overview of the economic impacts of price-based and investment-focused climate policies in developing countries but do not focus on employment impacts.

The modeling approach: MINDSET

The Model of Innovation in Dynamic Low-Carbon Structural Economic and Employment Transformations, MINDSET is designed to analyze short- and medium-term sectoral output, labor demand, and carbon-related cross-border competitiveness effects of climate and development policy mixes. The current version of MINDSET is a demand-driven global macroeconomic model with exogenous response functions. Based on the global GLORIA MRIO database², the model supports multi- or single-country policy scenario analyses, including impact analyses from third-party policies.³ Policy scenarios include the sectoral climate and development measures such as carbon pricing, border adjustments, income and payroll tax reforms, public spending and investment, electricity sector reforms (jointly with energy sector models), and climate damages (see Table 1 for detailed model inputs and outputs).

Table 1: Granular policy & climate scenarios modeled simultaneously for 120 sectors & 164 countries:

- a) **Carbon prices:** Commodity or carbon tax, subsidy, and tariffs, differentiated by commodity, sector, and country
- b) **Carbon Border Adjustment Mechanism (CBAM):** domestic carbon prices are complemented by border adjustments for user-defined energy-intensive and trade-exposed sectors, differentiated by sector and country for various global collaboration scenarios
- c) **Labor tax reforms,** differentiated by skill level
- d) **Income tax reforms**
- e) Sectoral **public spending policies,** cross-funded or additional
- f) Sectoral **public investment,** cross-funded or additional
- g) **Climate damages**
- h) **Clean energy investment**

*Model outputs inform various **metrics of interest** to support policy design:*

- Sectoral **output multipliers** of tax and spending policies,
- Sectoral **labor demand,** or employment multipliers of tax and spending policies
- **Distributional employment and consumption effects,** differentiated by income, skill level, occupation type and, spatially, by subnational province
- Sectoral **tax revenue**

² We used release 057 of the GLORIA global environmentally-extended multi-region input-output (MRIO) database (Lenzen et al. 2021), constructed in the Global MRIO Lab (Lenzen et al. 2017).

³ The model has been used in several climate-informed country analytics, such as the [Bosnia and Herzegovina Country Economic Memorandum](#), CCDRs in the [Philippines](#), [Peru](#), and [Bangladesh](#), as well as country operations, such as a Morocco Climate Operation PforR.

- Sectoral consumer **price differences & consumption distributional effects**
- **Competitiveness** effects and differences in sectoral **import and export volumes**

The main transmission mechanism translates climate policies into price and demand changes and simulates the economy's response to different climate policy scenarios. The results then are typically given as scenario differences, such as percentage difference of employment, output, or other economic indicators. The approach needs assumptions regarding the responses of industries and households to the scenario's price level and labor tax level. These responses are determined by the respective elasticities, with own- and cross-price elasticities for energy carriers as intermediate inputs to production and price elasticities of final demand from households and Governments. The same holds true for labor taxes.

MINDSET combines the strengths of input-output analysis – namely, short- to medium-term economic responses to exogenous demand differences, accounting for all multiplier effects from intermediate demand along the value chains in a consistent framework – with responses to exogenous price impacts. It connects all 164 countries in the dataset to each other through bilateral trade flows, thereby capturing trade effects from intermediate and final demand changes in one country on its trading partners. It estimates indirect and supply-chain impacts, as well as price-induced differences in international trade. The model's core elements are the interindustry relations given by intermediate use at basic prices by economic sector and final demand vectors by country connected by respective bilateral trade. The model does not capture labor flows into or out of the labor force or unemployment, nor between countries.

Introducing endogenous adjustments of production, consumption, and trade into the input-output framework, the model predicts policy-induced shifts across all sectors and all countries or regions. The model estimates indirect and supply-chain impacts, as well as price-induced changes in international trade. Coverage is disaggregated into 120 sectors, including 10 energy sectors, across 164 countries and regions, of which 133 are included here. The model is parametrized with the various sector-country-(fuel)-specific elasticities, including energy efficiency and fuel switching, household income and demand, trade, and output-induced investment (and converters), and output-employment elasticities. Augmentation with national labor force microdata allows for distributional labor demand analyses across income strata, skill levels, occupation types, and subnational provinces, providing relevant information for active labor market and reskilling interventions, as well as sectoral or regional government support funds.

Caveats

MINDSET does not assume that all available economic capacity is utilized. Instead, the level of effective demand determines the use of factors of production in the economy, including labor. In the input-output tradition, prices in the model are mostly fixed, assuming that firms draw upon available capacity. Thus, the supply level of material and labor inputs matches overall demand. This assumption is well aligned with empirical data. For example, European Union countries utilize just under 80 percent of their production capacity. This share drops to as low as just above 60 percent for Europe's upper-middle-income countries (European Commission 2024), and can be expected to be even lower in low- and lower-middle-income countries. The assumption of additionality of investments (as opposed to crowding out) and unconstrained labor supply (involuntary unemployment) may overstate the expansionary effects of climate policies. The model, therefore, complements the Computable General Equilibrium (CGE) approach. That said, MINDSET may be more appropriate than GE models for identifying transition im-

pacts and informing targeted complementary policy measures that could reduce short-term frictions. It also serves well for studying employment impacts, where labor market frictions or idle resources could have important socioeconomic consequences.

Depending on the structure of a country's economy, the assumption of additionality of unconstrained labor supply (involuntary unemployment) and investments (as opposed to crowding out) may slightly overstate the expansionary effects of climate mitigation policies. However, in countries with a large share of informal labor or unemployment, the assumption of strong constraints from the labor supply side (such as those captured in CGE models) may understate the expansionary macro effects of tax cuts or spending increases. Suppose a model assumes that development investments, such as in the health sector, have to crowd out investments from other areas of the economy (as assumed in CGE models). In that case, this will likely lead to negative GDP effects of climate or development policies. Because MINDSET does not assume that the economy is at full capacity in the baseline, i.e., not at the efficiency frontier, the model further does not automatically allow prices to adjust. Thus, when demand increases (e.g., in the health sector), prices are not immediately pushed up.

One area that is not yet considered well in model-based analysis is the mobility of labor between sectors, regions, and countries (Spencer et al. 2018). Models either assume perfect labor mobility (equilibrium tools) or constraints based on historical trends (macro-econometric models). However, the scale of structural change in the low-carbon transition could run up against skill- and location-based constraints that have not previously been seen. Methods to assess these constraints are being developed (e.g., Mealy and Coyle 2022) and will be integrated into a future version of MINDSET. However, the policy scenarios analyzed in this paper do not trigger fundamental structural shifts in labor demand, such that labor supply to meet the demand is unlikely to be constrained by labor market dynamics.

Scenario description

Table 2: Scenario overview

	Carbon Tax ⁴	Revenue re-cycling	Additional spending	Description
Gov	\$80/tCO ₂	Public Spending	None	65% of the revenues are used for public <i>investment</i> in education and healthcare (e.g., construction and machinery procurement of hospitals and schools) and 35% for public <i>spending</i> on education and healthcare (hiring more doctors, nurses, and teachers ⁵).
PR	\$80/tCO ₂	Payroll Tax Cut	None	Revenues are used to reduce payroll taxes, with half of the revenues reducing labor costs for employers and the other half going to households as transfers.
PR+RE	\$80/tCO ₂	Payroll Tax Cut	Clean energy investment	100% of revenues is used to reduce payroll taxes. Investment in renewable energy and energy efficiency needed to close the gap to net-zero GHG emission electricity generation is not funded by the carbon tax, but rather by exogenous sources.

Three scenarios have been implemented in the model and are compared to a business-as-usual scenario without additional climate policy in the simulation horizon, i.e., 2030. The overarching driver of people's and firms' behavior is the introduction of a global carbon tax. The policy scenarios are identical across all countries to facilitate comparison across countries and to illustrate the drivers of different economic outcomes. The same holds for the suggested revenue recycling schemes. Scenarios are identical across countries in terms of the percentages of the revenue allocated to the different purposes. In terms of levels, however, they differ according to the respective revenues collected in each country.

The payroll tax cut scenarios try to capture the double dividend hypothesis, taxing fossil fuels and un-taxing jobs. Although this is somewhat simplistic, it illustrates how reducing labor costs in combination with carbon taxes leads to a shift from energy to labor as input factors of production. Half of the revenues from the carbon tax are used to lower labor costs, the other half is transferred to households, who spend it according to their existing expenditure patterns.

Alternatively, revenue can be spent investing in health and education infrastructure and people. The health and education sectors are important for human capital development and, in most countries, underfinanced. Infrastructure is lacking, including in high-income countries, and is only gradually adapting

⁴ The suggested level of \$80/tCO₂ lies within the corridor concluded by the High-Level Commission on Carbon Prices, Carbon Price Leadership Coalition, 2017.

⁵ Note that given the timeframe of the analysis, secondary effects from having a healthier and better educated workforce are not included in the reported analysis.

to climate change. Schools need to be climate-resilient so that education is not hampered by extreme weather events. Schools can be turned into shelters in their neighborhoods, too⁶ (Feinstein and Mach 2020). Hospitals and clinics need infrastructure upgrades to be resilient and able to cope with extreme weather events or slow-onset changes. The public spending scenario picks up this idea and uses 65% of the revenue from the carbon tax for investment in resilient hospitals and schools. The remaining 35% is spent on staff, which could, for instance, include hiring climate health and education experts.

Additional mitigation measures are needed because taxing carbon does not induce changes away from fossil fuel use to the extent needed to achieve climate targets. The gap between a carbon tax rate acceptable to many countries and implementable and the necessary push for moving out of fossil fuels must be bridged by other measures. The scenario PR+RE assumes additional investment in clean energy, which has been determined by identifying the gap in renewable energy installations for electricity generation and efficiency increases under the IEA's net zero and current policies scenarios. This gives a country-specific investment pathway that will unfold the respective economic impacts under the respective country's economic structure.

The results are driven by the interplay between price increases stemming from the carbon tax for all economic agents, income increases from income support and additional jobs, and demand increases from additional household expenditures and investment in clean energy. Higher prices for fossil fuels induce technology shifts for industries, trying to replace expensive inputs with less expensive ones or cutting back production. Households' budgets are similarly impacted by direct price increases of fossil fuels and indirect price increases of goods and services using fossil fuels as inputs. Payroll tax cuts reduce the price of labor and are expected to lead to more jobs across the economy. Higher employment levels yield higher income and more demand for domestically produced and imported goods. Investment in renewable energy yields additional demand for goods and services, both during the installation phase and later for operation and maintenance. If the carbon tax is applied across all countries, the terms of trade are not largely affected, and trade effects occur from increased demand for renewable energy goods, reduced consumption of fossil fuels, and increased consumption and investment goods from additional spending.

⁶ Examples are the COOLSCHOOLS project <https://www.uoc.edu/en/news/2022/072-climate-shelters-schools>, or the activities of the World Bank, who has between 2014 and 2023, supported 35 countries through projects that have benefited over 121 million students in around 564,000 schools and informed the design and implementation of more than \$3.1 billion in World Bank-financed school infrastructure operations. <https://www.worldbank.org/en/results/2024/03/08/building-safer-and-more-resilient-schools-in-a-changing-climate>

Results

The impacts on economic and employment outcomes differ across regions, sectors, country income levels, and countries with different resource endowments. The results are presented as relative differences to the baseline. Indicators reported are relative differences in total employment, GDP, and sectoral employment (for the 10 economic sectors listed in **Error! Not a valid bookmark self-reference.**), and sectoral output. The results are grouped into seven regions, four income levels, and being a net crude or refined oil importer or exporter. The sample contains 46 countries in HIC, 11 in LIC⁷, 37 observations in LMIC, and the remaining 38 in UMIC, while 15 countries are in the Eastern Pacific region, 47 in Europe and Central Asia, 20 in Latin America and the Caribbean, 16 in MENA and 2 in North America, 2 in South Asia and 28 in Sub-Saharan Africa⁸.

Table 3: Result dimensions⁹

Economic sectors	Regions	Resources	Income Groups
•A. Agriculture; forestry and fishing •B. Mining and quarrying •C. Manufacturing •D. +E. Utilities •F. Construction •G.-N. Services •O. Public administration and defense; compulsory social security •P. Education •Q. Human health and social work activities •R.+S. Other	• East Asia & Pacific • Europe & Central Asia • Latin America & Caribbean • Middle East & North Africa • North America • South Asia • Sub-Saharan Africa	•Oil importer •Oil exporter •Petroleum importer •Petroleum exporter	•High Income (HIC) •Upper Middle Income (UMIC) •Lower Middle Income (LMIC) •Low Income (LIC)

Source: Own compilation

Overview global results

The overall employment effect of all scenarios is small but positive compared to the reference. Under the PR Scenario, the median value across all countries for the relative difference in employment is 0.2%, meaning employment is at least 0.2% higher than in the baseline for half of all countries. Albeit small, this is significant because it indicates that positive employment effects prevail. In 85% of all countries the impact of the carbon tax plus payroll tax reduction scenario (PR) is positive. Combining this with additional investment in renewable energy (under the PR+RE scenario) drives the median impact to 0.6%, and spending revenues on health and education (as posited under the Gov scenario) leads to 0.3% additional jobs. Lower middle-income countries experience slightly larger positive employment effects under the combined Pr+RE scenario, and the GOV scenario, partly because productivity is still low, yielding many jobs from additional output. However, revenue recycling through payroll tax reductions alone has a

⁷ Venezuela is currently not assigned an income group

⁸ Note that MINDSET contains 163 countries. The simulation results in this report cover 133 countries. The difference in country count is mainly due to country size and FCV status.

⁹ The classification of sector has been aggregated based on main ISIC sectors to make the result dimensions more manageable.

low effect on LICs. High-income countries benefit similarly from additional spending in renewable energy and see the highest effect from government spending compared across country income classes.

Table 4: Impacts of different scenarios on employment, relative difference to reference.

	<i>PR</i>	<i>PR+RE</i>	<i>Gov</i>
<i>All</i>	0.2%	0.6%	0.3%
<i>LIC</i>	0.3%	0.7%	0.1%
<i>LMIC</i>	0.2%	0.7%	0.3%
<i>UMIC</i>	0.2%	0.6%	0.3%
<i>HIC</i>	0.2%	0.6%	0.4%

Source: MINDSET model results.

GDP globally and by income group is relatively higher under all scenarios. GDP effects are highest under the payroll tax reduction scenario combined with investment in renewable energy technologies. LICs benefit strongly under this scenario especially if they are dependent on fossil fuel imports, because increased RE deployment reduces these imports. Relative to jobs impacts, GDP increases more strongly in HICs. The payroll tax reduction alone yields a lower GDP difference than the scenario with government spending on health and education plus household income support.

Table 5: Impacts of different scenarios on GDP, relative difference to reference.

	<i>PR</i>	<i>PR+RE</i>	<i>Gov</i>
<i>All</i>	0.4%	1.3%	0.6%
<i>LIC</i>	0.2%	1.0%	0.1%
<i>LMIC</i>	0.3%	1.2%	0.3%
<i>UMIC</i>	0.4%	1.3%	0.8%
<i>HIC</i>	0.5%	1.3%	0.8%

Source: MINDSET model results.

Output effects fall somewhere in the middle between employment effects and GDP effects, except for high-income countries. In high-income countries, the output effect is highest and most detached from the impact on jobs. The next section takes a deep dive into disaggregated sector effects and provides a structural explanation for these observations. Under the government spending plus household income support scenario, LIC cannot benefit as much in output because a large share of the additional demand will be met by imports. Sectoral disaggregation will illustrate this point further.

Table 6: Impacts of different scenarios on output, relative difference to baseline.

	<i>PR</i>	<i>PR+RE</i>	<i>Gov</i>
<i>All</i>	0.2%	1.3%	0.6%
<i>LIC</i>	0.3%	1.1%	0.1%
<i>LMIC</i>	0.1%	1.1%	0.3%
<i>UMIC</i>	0.2%	1.2%	0.8%
<i>HIC</i>	0.4%	1.4%	0.8%

Source: MINDSET model results.

Economic and employment effects by region and sector

The differences in economic outcomes are equally dissimilar across regions as across income groups. Median employment impacts under the payroll tax reduction plus renewable energy investment scenario by region are small for East Asia Pacific and North America. The largest variation in outcomes is observed across MENA countries, while Sub-Saharan African countries cluster much narrower around their median value of a little more than .3%. Europe and Central Asia, too, show the results more concentrated around over 1% relative difference in employment under this scenario (**Error! Not a valid bookmark self-reference.**). High-income countries show a narrower range of employment impacts, with a standard error of 0.11% around an employment impact of .2%, while low and lower-middle-income countries have a larger variation in their respective results. For countries in regions and income groups with fewer variations, turning to their respective peers gives more information regarding their expected performances. In MENA and in LICs and LMICs, peer countries from the same group can exhibit very different results and may lead to invalid expectations.

Table 7: Median values and confidence intervals for the PR scenario.

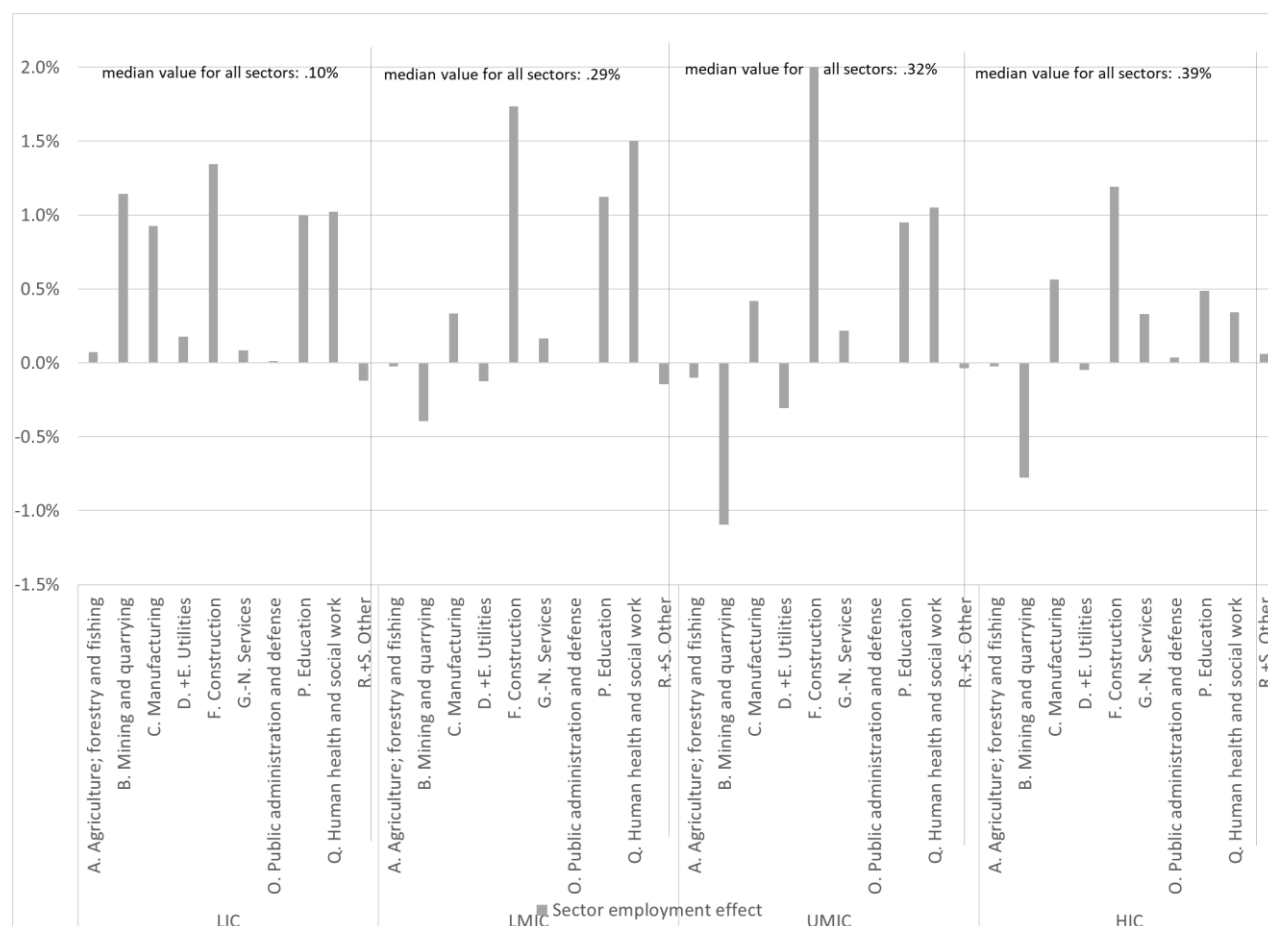
	Median	[95% conf. interval]	
Region			
East Asia & Pacific	0.31%	0.16%	0.42%
Europe & Central Asia	0.24%	0.19%	0.29%
Latin America & Caribbean	0.25%	0.18%	0.35%
Middle East & North Africa	0.06%	-0.02%	0.29%
North America	0.21%	0.10%	0.32%
South Asia	0.31%	0.12%	0.50%
Sub-Saharan Africa	0.31%	0.13%	0.41%
Income Group			
Low Income	0.34%	0.01%	0.50%
Lower Middle Income	0.23%	0.15%	0.33%
Upper Middle Income	0.22%	0.11%	0.32%
High Income	0.22%	0.18%	0.29%

Source: MINDSET model results.

Recycling the carbon tax revenues into spending and investing in education and health as well as in income support households (Gov scenario) creates positive employment impacts in the targeted sectors and the whole economy. **Error! Not a valid bookmark self-reference.** shows the relative differences under this scenario compared to employment in the respective sector under the baseline, grouped by income group. In all income groups, targeted investment and spending for the education and health sectors are reflected in additional employment. For LICs, additional employment in the health and education sector amounts to 1% of baseline sectoral employment. Additional investment in these sectors triggers construction activities and is reflected in almost 1.2% more jobs in construction. The construction sector is one of the largest non-agricultural sectors in many LIC economies. Manufacturing benefits from additional demand because many households have more income due to transfers and higher employment. Services benefit along the value chain but do not see strong effects under this scenario. The uptake of the mining sector is related to quarrying along the value chain of the construction sector. Moving to the other income groups, this effect is more than offset by reductions in fossil fuel mining because of

the carbon tax and the resulting impact on mining is negative; most prominently in UMICs with more than 1% job losses. Albeit these losses are compensated on average by all other positive impacts, the effect on mining regions can be quite severe. UMICs also see the largest negative impacts on electricity generation based on fossil fuels, whereas HIC already have parts of this industry devoted to cleaner energy options and not being as much negatively affected by the carbon tax.

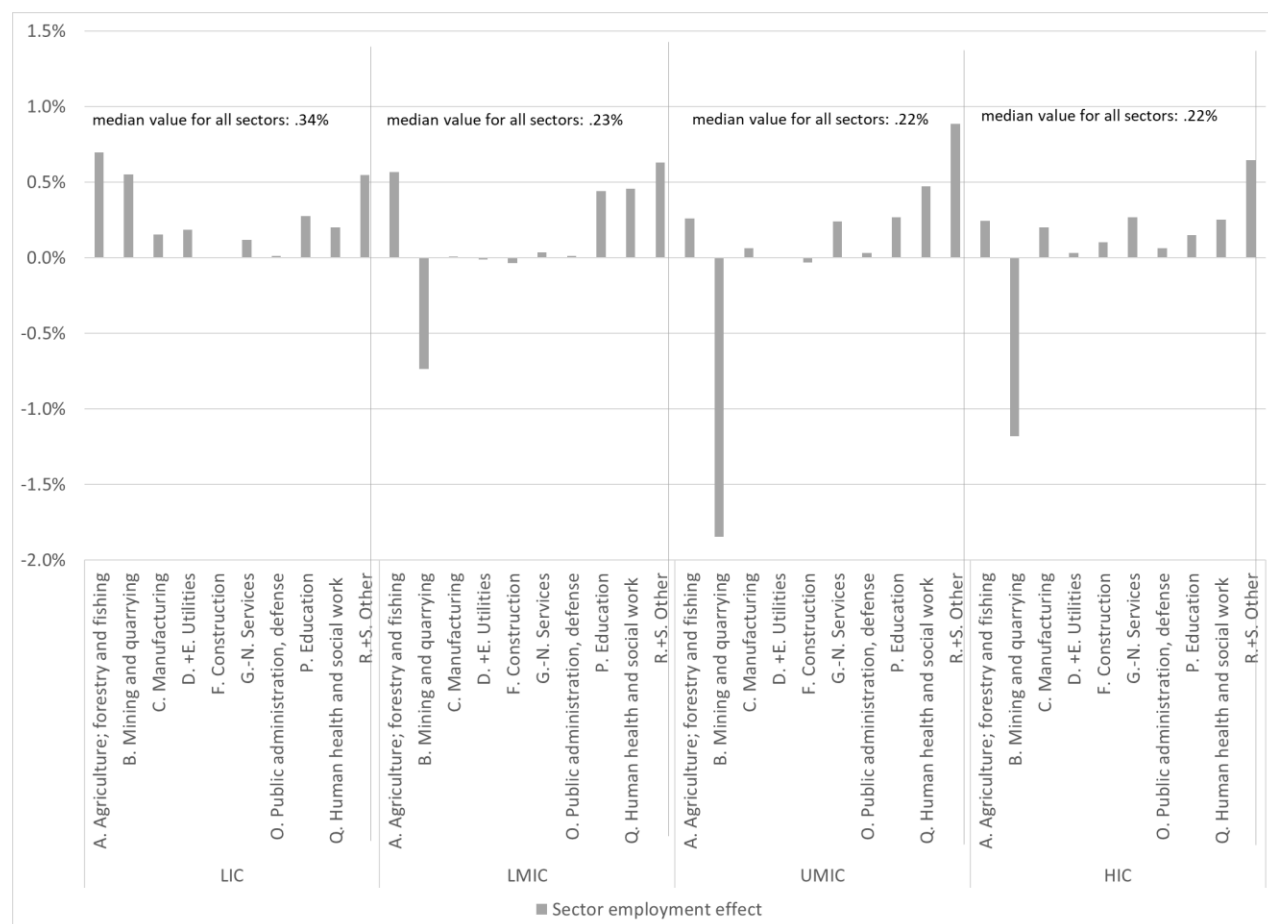
Figure 1: Employment effects by sector and income group, in relative differences of sectoral employment to baseline under 'Gov' scenario



Source: MINDSET model results.

Under the payroll tax cut scenario, effects are smaller and more evenly distributed across sectors. In Figure 2, the manufacturing and service sectors benefit more from this scenario than from the previously discussed redistribution scheme. Health and education also see additional employment under this scenario, partly from improved labor costs and partly from increased incomes and people's willingness to spend more on health and education. The construction sector sees fewer impacts, because the targeted investment in health and education facilities is missing. The pattern in energy and mining is comparable to the scenario previously discussed; negative impacts in electricity generation are more relevant in more developed countries due to price increases, loss of demand, and less additional demand from increased economic activities compared to the scenario Gov. The largest positive effects are observed in some services and hiring by households, as shown in Figure 2.

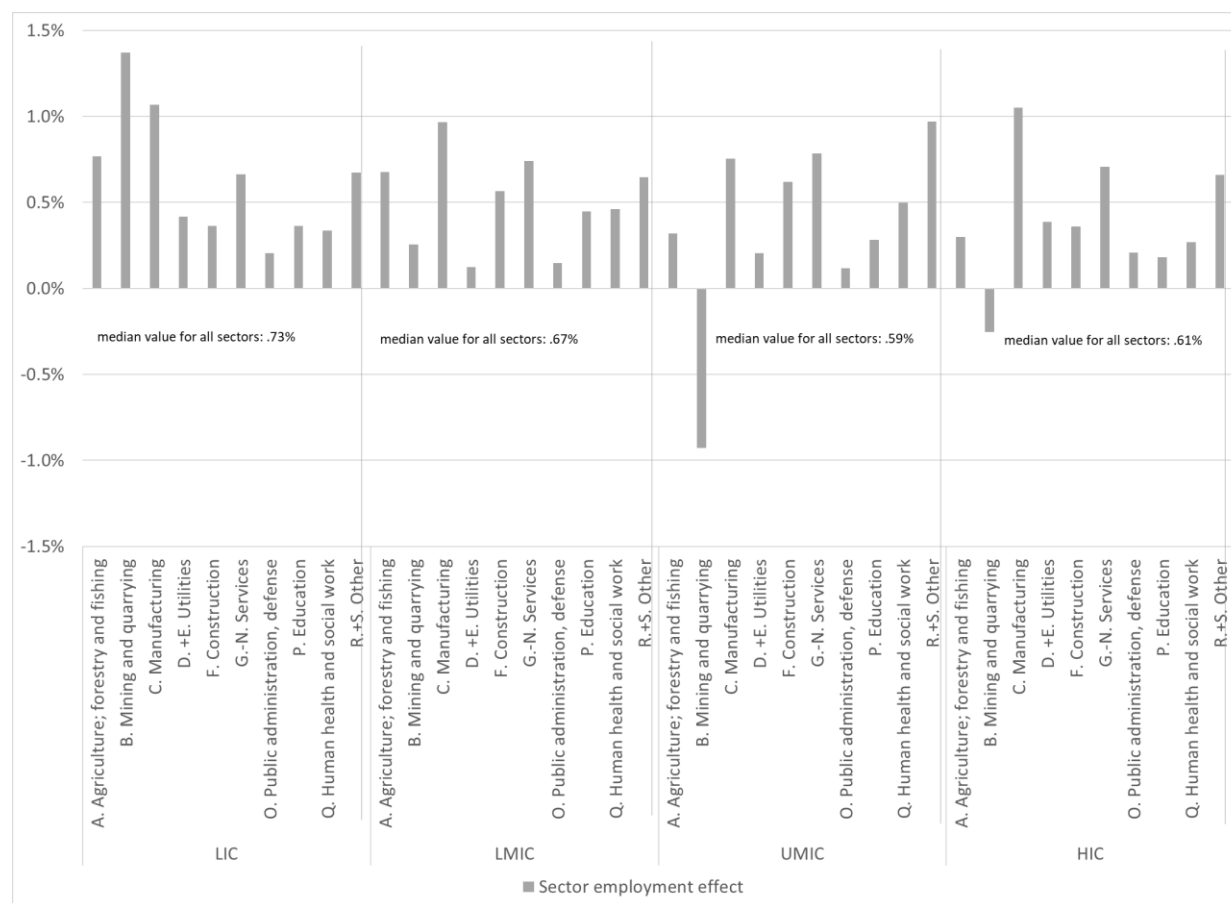
Figure 2: Employment effects by sector and income group, in relative differences of sectoral employment to baseline under scenario PR

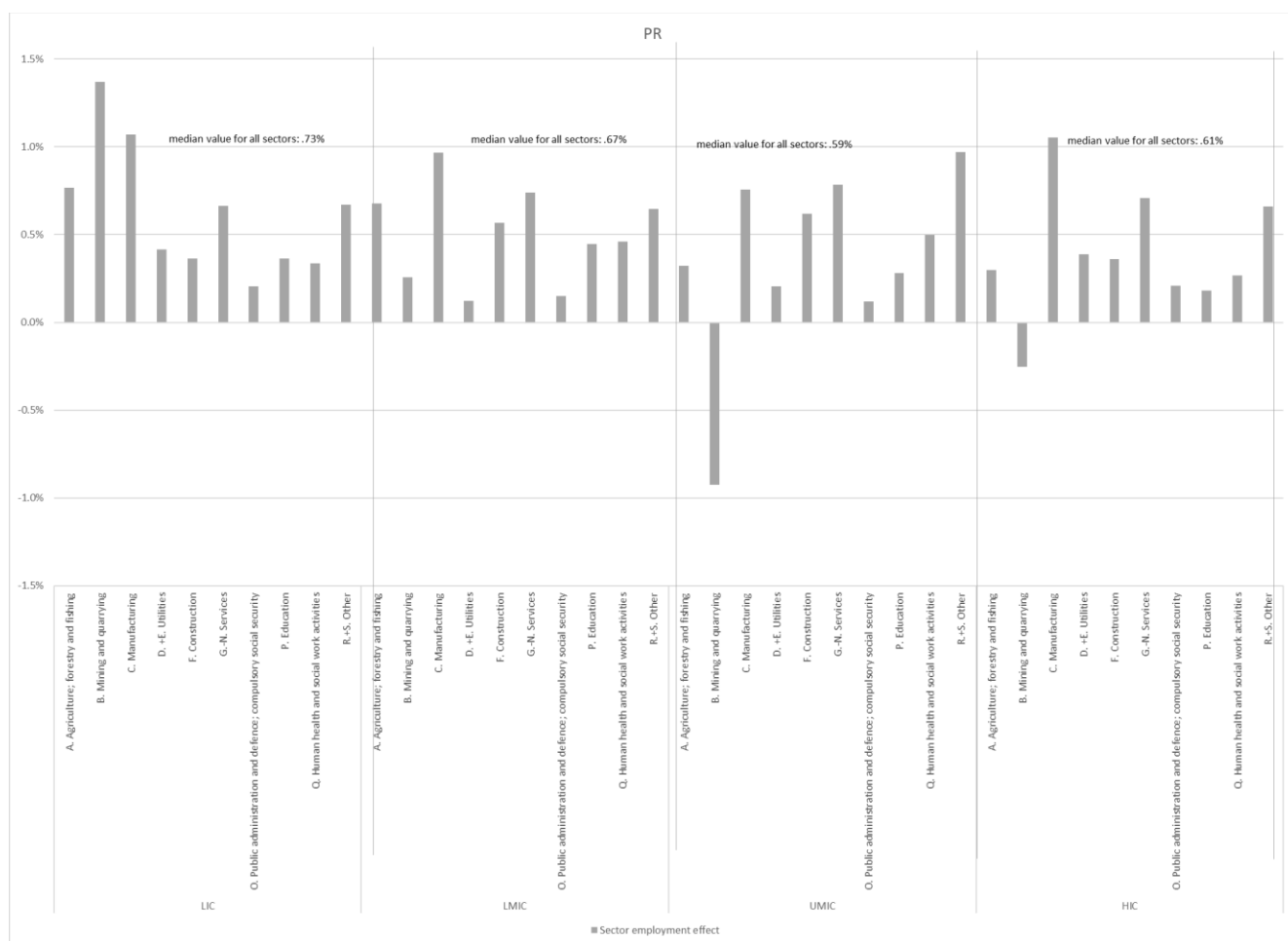


Source: MINDSET model results.

Combining clean energy investment with the payroll tax reduction (scenario PR+RE) leads to the highest effects. Additional employment is seen across most sectors, with the construction sector further benefiting from clean energy investment. Investment in renewable energy, as well as increased energy efficiency, creates additional demand for output from the construction sector. HICs and UMICs see additional jobs in the manufacturing sector, which are above the average across all sectors. This may be due to the demand for clean energy equipment and technologies, which are meeting additional global demand. Services respond to additional intermediate demand and lower labor costs, showing above-average increases in labor demand.

Figure 3: Employment effects by sector and income group, in relative differences of sectoral employment to baseline under scenario PR+RE





Source: MINDSET model results.

Summary of employment effects across countries

The effect of the outlined revenue-neutral carbon tax schemes can differ across countries, depending on their economic structure and income level. Figure 4 visualizes the employment and GDP changes for payroll tax cut and government expenditure scenarios in selected countries. Middle-income countries such as in East Asia and Pacific (Indonesia, Vietnam) can benefit more from revenue-neutral carbon tax compared to high-income countries such as in European countries (Germany, Switzerland) for either scenario (also see Table 4 and 5). In more detailed investigation, certain middle-income countries with lower labor productivity, such as Indonesia and Ecuador, are benefitting more from payroll tax cut, where more services jobs can be created from the productivity increase. The payroll tax reduction alone can absorb more labor in these countries, although targeted government spending can increase the GDP since the healthcare and education sectors are high-value-added sectors.

The global results from this rather standardized exercise would need to be complemented by a more detailed analysis of job quality, informality, skill needs and gender distribution to paint a more complete picture. Additional job impacts from consumption demand, for instance, often occur in sectors that are characterized by high shares of job informality, unwaged jobs, or family labor in the agriculture

sector. Construction, where large impacts are observed, likewise exhibits high shares of informality. Climate policies need to be complemented by the respective labor oriented policies to have the green transition lead to more and better jobs.

In general, oil exporting countries (in form of crude oil or petroleum products) with higher income level can benefit more from revenue-neutral carbon tax than net oil importers. In major oil producing countries in Middle East such as Saudi Arabia, Oman, and Iran, revenue-neutral carbon tax creates a positive impact on employment and GDP. Meanwhile, oil producing countries in other regions such as Brunei, Turkmenistan, and Nigeria, revenue-neutral carbon tax does not generate as positive an impact as observed in the Middle East region. Since Middle Eastern countries have both more efficient technology and more diversified economy compared to other oil exporting countries, the impact of carbon tax on the economy is relatively lower and the high tax revenue obtained could then be used to develop the existing non-petroleum industries in these countries, particularly if they have more diverse industry structure.

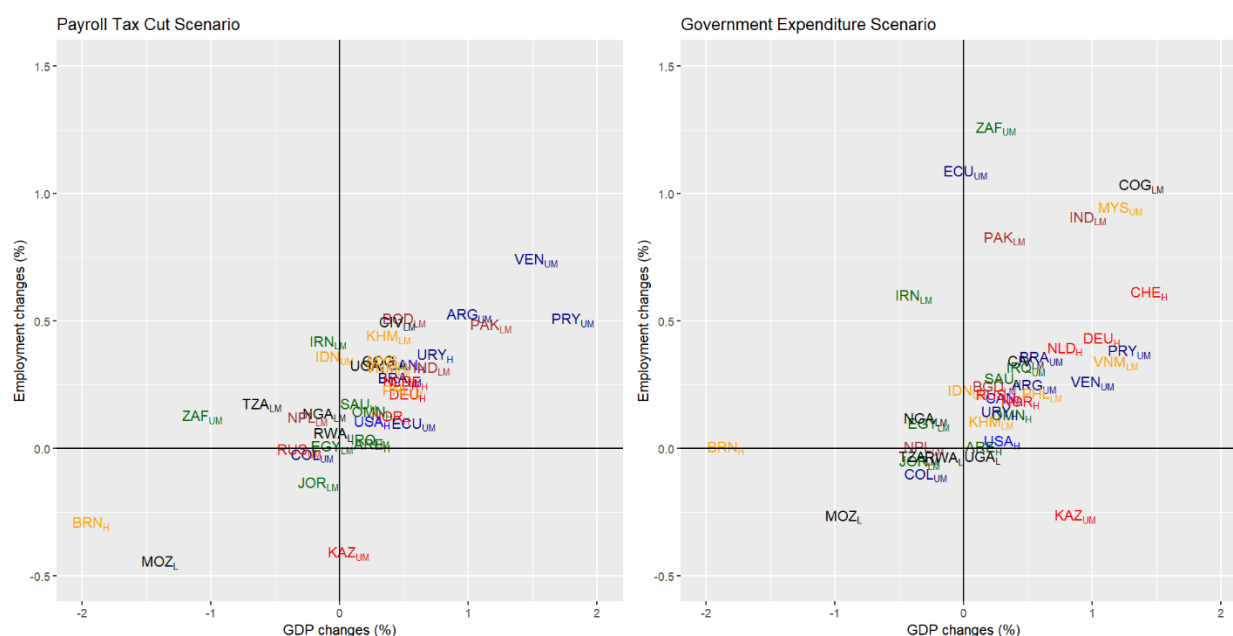
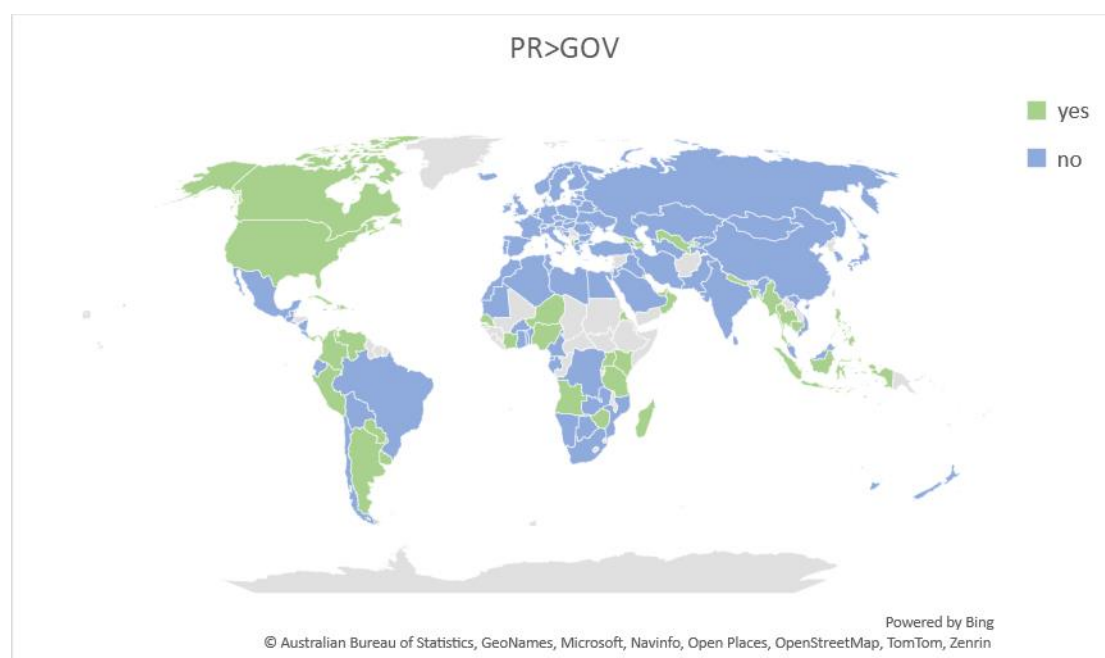


Figure 4: Impacts of revenue-neutral carbon tax in payroll tax cut (PR) and government expenditure (GOV) scenarios on GDP and employment, relative difference to reference. The color represents the country region based on World Bank classification.

Payroll taxes plus renewable energy investment result in greater economic benefits than government spending with income support in most countries. This confirms the finding in the literature, as Tilimisa 2022 also points out in the literature review on the benefits of carbon taxation. Most countries fare better under the increased health and education spending and investment in making the health sector more resilient, by investing in buildings, early warning systems etc. The mere effects from recycling the revenue to reduce distortionary taxes, in this case payroll taxes, do not exceed the benefits from the targeted government spending. Figure 4 displays a map indicating where employment impacts under the PR scenario are larger than under the Gov scenario (“yes”, denoted in green) or not. **Error! Reference source not found.** poses – and answers – the same question for the PR+RE scenario.

Figure 5: Comparison of effects across countries. Comparing the PR scenario with the Gov scenario



Source: MINDSET model results.

Adding renewable energy investment changes the picture. Notably, relative differences between scenario effects are not represented in (Figure 5), and for several countries, the economic effects are of very similar size. However, the combined payroll tax reduction and investment scenario does better than the revenue recycling into targeted government spending for all countries.

Conclusions and outlook

This analysis provides very detailed results for countries to inform their understanding and expectations for different carbon tax schemes. The results illustrate how different countries respond to a stylized and generalized set of climate policies. The combination of carbon taxes and revenue recycling through payroll tax reduction seems to be the most efficient and the most beneficial in terms of additional jobs. It removes negative externalities of both emissions and distortionary taxes, and hence is theoretically appealing and yielding a double dividend. To more fully understand the implications and factors which are driving the results, a wider net on explaining factors could be cast. Potential indicators could be population or area size.

This analysis is part of a broad research agenda on the employment implications of decarbonization and green transition (that is, to more climate-resilient and sustainable economies). Future research will explore emerging policy questions of relevance to policymakers, such as revenue recycling schemes with transfers to only lower-income quantiles, to better target the poor and vulnerable groups. The literature on fossil fuel subsidy reform might provide some compelling examples (IISD 2019; World Bank 2022).

Deep dives into subsets of countries would allow more analysis of skills needed and existing, gender balance of results and other labor market aspects. Global results allow cross country comparison with peers or aspirational peers. Country deep dives then allow the analysis of drivers under different scenarios and a better understanding of the transmission pathways of employment effects in the economy. The

MINDSET model has been used in country specific analyses, such as (Lehr and Pollitt 2024). Further scenarios, as outlined above would be of interest in the decision process for the urgently need more stringent climate policies for both mitigation and adaptation.

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